

Alcohol use disorder	Diagnosed using criteria for substance use disorder. Complications: vitamin B ₁ (thiamine) deficiency, alcoholic cirrhosis, hepatitis, pancreatitis, peripheral neuropathy, testicular atrophy. Treatment: naltrexone (reduces cravings; avoid in liver failure), acamprosate (contraindicated in renal failure), disulfiram (to condition the patient to abstain from alcohol use). Support groups such as Alcoholics Anonymous are helpful in sustaining abstinence and supporting patient and family.
Wernicke-Korsakoff syndrome	Results from vitamin B ₁ deficiency. Symptoms can be precipitated by administering dextrose before vitamin B ₁ . Triad of confusion, ophthalmoplegia, ataxia (Wernicke encephalopathy). May progress to irreversible memory loss, confabulation, personality change (Korsakoff syndrome). Treatment: IV vitamin B ₁ (before dextrose).

► PSYCHIATRY—PHARMACOLOGY

Psychotherapy

Behavioral therapy	Teaches patients how to identify and change maladaptive behaviors or reactions to stimuli (eg, systematic desensitization for specific phobia).
Cognitive behavioral therapy	Teaches patients to recognize distortions in their thought processes, develop constructive coping skills, and ↓ maladaptive coping behaviors → greater emotional control and tolerance of distress (eg, recognizing triggers for alcohol consumption).
Dialectical behavioral therapy	Designed for use in borderline personality disorder, but can be used in other psychiatric conditions as well (eg, depression).
Interpersonal therapy	Focused on improving interpersonal relationships and communication skills.
Motivational interviewing	Enhances intrinsic motivation to change by exploring and resolving ambivalence. Used in substance use disorder and weight loss.
Supportive therapy	Utilizes empathy to help individuals during a time of hardship to maintain optimism or hope.

Central nervous system stimulants	Methylphenidate, dextroamphetamine, methamphetamine, lisdexamfetamine.
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MECHANISM	↑ catecholamines in the synaptic cleft, especially norepinephrine and dopamine.
CLINICAL USE	ADHD, narcolepsy, binge-eating disorder.
ADVERSE EFFECTS	Nervousness, agitation, anxiety, insomnia, anorexia, tachycardia, hypertension, weight loss, tics, bruxism.

Antipsychotics

Typical (1st-generation) antipsychotics—haloperidol, pimozide, trifluoperazine, fluphenazine, thioridazine, chlorpromazine.

Atypical (2nd-generation) antipsychotics—aripiprazole, asenapine, clozapine, olanzapine, quetiapine, iloperidone, paliperidone, risperidone, lurasidone, ziprasidone.

MECHANISM	Block dopamine D ₂ receptor (↑ cAMP). Atypical antipsychotics also block serotonin 5-HT ₂ receptor. Aripiprazole is a D ₂ partial agonist.
CLINICAL USE	Schizophrenia (typical antipsychotics primarily treat positive symptoms; atypical antipsychotics treat both positive and negative symptoms), disorders with concomitant psychosis (eg, bipolar disorder), Tourette syndrome, OCD, Huntington disease. Clozapine is used for treatment-resistant psychotic disorders or those with persistent suicidality (cloze to the edge).
ADVERSE EFFECTS	Antihistaminic (sedation), anti-α_1-adrenergic (orthostatic hypotension), antimuscarinic (dry mouth, constipation) (anti-HAM). Use with caution in dementia. Metabolic: weight gain, hyperglycemia, dyslipidemia. ↑ risk with clozapine and olanzapine (obesity). Endocrine: hyperprolactinemia → galactorrhea, oligomenorrhea, gynecomastia (↓ activity in the tuberoinfundibular pathway). Cardiac: QT prolongation. Neurologic: neuroleptic malignant syndrome. Ophthalmologic: chlorpromazine—corneal deposits; thioridazine—retinal deposits. Clozapine—agranulocytosis (monitor WBCs clozely), seizures (dose related), myocarditis, hypersalivation. Extrapyramidal symptoms (↓ activity in the nigrostriatal pathway)—ADAPT: - Hours to days: Acute Dystonia (muscle spasm, stiffness, oculogyric crisis). Treatment: benztropine, diphenhydramine. - Days to months: - Akathisia (restlessness). Treatment: β-blockers, benztropine, benzodiazepines. - Parkinsonism (bradykinesia). Treatment: benztropine, amantadine. - Months to years: Tardive dyskinesia (chorea, especially orofacial). Treatment: benzodiazepines, botulinum toxin injections, valbenazine, deutetrabenazine.
NOTES	Lipid soluble → stored in body fat → slow to be removed from body. Typical antipsychotics have greater affinity for D₂ receptor than atypical antipsychotics → ↑ risk for hyperprolactinemia, extrapyramidal symptoms, neuroleptic malignant syndrome. High-potency typical antipsychotics: haloperidol, trifluoperazine, pimozide, fluphenazine (Hal tries pie to fly high)—more neurologic adverse effects (eg, extrapyramidal symptoms). Low-potency typical antipsychotics: chlorpromazine, thioridazine (cheating thieves are low)—more antihistaminic, anti-α_1-adrenergic, antimuscarinic effects.

Lithium

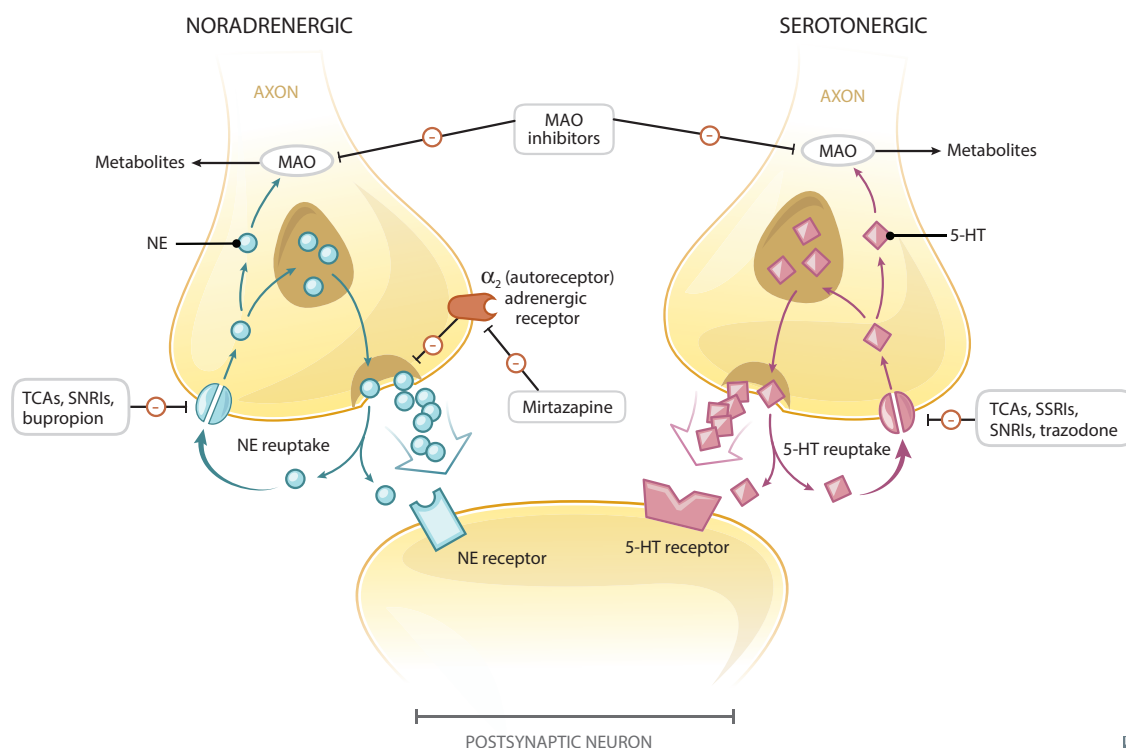
MECHANISM	Affects neurotransmission (↓ excitatory, ↑ inhibitory) and second messenger systems (eg, G proteins).
CLINICAL USE	Mood stabilizer for bipolar disorder; treats acute manic episodes and prevents relapse.
ADVERSE EFFECTS	Tremor, hypothyroidism, hyperthyroidism, mild hypercalcemia, polyuria (causes nephrogenic diabetes insipidus), teratogenesis (causes Ebstein anomaly). Narrow therapeutic window requires close monitoring of serum levels. Almost exclusively excreted by kidneys; most is reabsorbed at PCT via Na ⁺ channels. Thiazides, ACE inhibitors, NSAIDs, and other drugs affecting clearance are implicated in lithium toxicity. LiTHIUM: Low T hyroid (hypothyroidism) Heart (Ebstein anomaly) Insipidus (nephrogenic diabetes insipidus) Unwanted M ovements (tremor)

Buspirone

MECHANISM	Partial 5-HT _{1A} receptor agonist.
CLINICAL USE	Generalized anxiety disorder. Does not cause sedation, addiction, or tolerance. Begins to take effect after 1–2 weeks. Does not interact with alcohol (vs barbiturates, benzodiazepines). I get anxious if the bus doesn't arrive at one , so I take buspirone .

Antidepressants

It normally takes 4–8 weeks for antidepressants to show appreciable effect.



Selective serotonin reuptake inhibitors

Fluoxetine, fluvoxamine, paroxetine, sertraline, escitalopram, citalopram.

MECHANISM	Inhibit 5-HT reuptake.
CLINICAL USE	Depression, generalized anxiety disorder, panic disorder, OCD, bulimia, binge-eating disorder, social anxiety disorder, PTSD, premature ejaculation, premenstrual dysphoric disorder.
ADVERSE EFFECTS	Fewer than TCAs. Serotonin syndrome, GI distress, SIADH, sexual dysfunction (anorgasmia, erectile dysfunction, ↓ libido), mania precipitation if underlying bipolar disorder.

Serotonin-norepinephrine reuptake inhibitors

Venlafaxine, desvenlafaxine, duloxetine, levomilnacipran, milnacipran.

MECHANISM	Inhibit 5-HT and NE reuptake.
CLINICAL USE	Depression, generalized anxiety disorder, diabetic neuropathy. Venlafaxine is also indicated for social anxiety disorder, panic disorder, PTSD, OCD. Duloxetine and milnacipran are also indicated for fibromyalgia.
ADVERSE EFFECTS	↑ BP, stimulant effects, sedation, sexual dysfunction, nausea.

Tricyclic antidepressants

Amitriptyline, nortriptyline, imipramine, desipramine, clomipramine, doxepin, amoxapine.

MECHANISM	TCAs inhibit 5-HT and NE reuptake.
CLINICAL USE	MDD, peripheral neuropathy, chronic neuropathic pain, migraine prophylaxis, OCD (clomipramine), nocturnal enuresis (imipramine).
ADVERSE EFFECTS	Sedation, α_1 -blocking effects including postural hypotension, and atropine-like (anticholinergic) adverse effects (tachycardia, urinary retention, dry mouth). 3° TCAs (amitriptyline) have more anticholinergic effects than 2° TCAs (nortriptyline). Can prolong QT interval. Tri-CyCliC's : C onvulsions, C oma, C ardiotoxicity (arrhythmia due to Na ⁺ channel inhibition); also respiratory depression, hyperpyrexia. Confusion and hallucinations are more common in older adults due to anticholinergic adverse effects (2° amines [eg, nortriptyline] better tolerated). Treatment: NaHCO ₃ to prevent arrhythmia.

Monoamine oxidase inhibitors

Tranylcypromine, phenelzine, isocarboxazid, selegiline (selective MAO-B inhibitor). (MAO takes pride in Shanghai).

MECHANISM	Nonselective MAO inhibition → ↑ levels of amine neurotransmitters (norepinephrine, 5-HT, dopamine).
CLINICAL USE	Atypical depression, anxiety. Parkinson disease (selegiline).
ADVERSE EFFECTS	CNS stimulation; hypertensive crisis, most notably with ingestion of tyramine. Contraindicated with SSRIs, TCAs, St. John's wort, meperidine, dextromethorphan, pseudoephedrine, linezolid (to avoid precipitating serotonin syndrome). Wait 2 weeks after stopping MAO inhibitors before starting serotonergic drugs (risk for serotonin syndrome) or stopping dietary restrictions (risk for tyramine induced hypertensive crisis).

Atypical antidepressants**Atypical Bees Make Tasty Vegan Venison.**

Bupropion	Inhibits NE and DA reuptake. Also used for smoking cessation. Adverse effects: stimulant effects (tachycardia, insomnia), headache, seizures in patients with bulimia and anorexia nervosa. ↓ risk of sexual adverse effects and weight gain compared to other antidepressants.
Mirtazapine	α_2 -antagonist (↑ release of NE and 5-HT), potent 5-HT ₂ and 5-HT ₃ receptor antagonist, and H ₁ antagonist. Adverse effects: sedation (which may be desirable in depressed patients with insomnia), ↑ appetite, weight gain (which may be desirable in underweight patients), dry mouth. Mirtazapine makes you munch more and move less.
Trazodone	Primarily blocks 5-HT ₂ , α_1 -adrenergic, and H ₁ receptors; also weakly inhibits 5-HT reuptake. Used primarily for insomnia, as high doses are needed for antidepressant effects. Adverse effects: sedation, nausea, priapism, postural hypotension. Think tra ZZZ o bone due to sedative and male-specific adverse effects.
Vilazodone	Inhibits 5-HT reuptake; 5-HT _{1A} receptor partial agonist. Used for MDD. Adverse effects: headache, diarrhea, nausea, anticholinergic effects. May cause serotonin syndrome if taken with other serotonergic agents.
Vortioxetine	Inhibits 5-HT reuptake; 5-HT _{1A} receptor agonist and 5-HT ₃ receptor antagonist. Used for MDD. Adverse effects: nausea, sexual dysfunction, sleep disturbances, anticholinergic effects. May cause serotonin syndrome if taken with other serotonergic agents.

Pharmacotherapies for smoking cessation

Nicotine replacement therapy	Binds to nicotinic ACh receptors. Aim to relieve withdrawal symptoms upon stopping smoking. Long-acting patch and short-acting products (ie, gum, lozenge) can be used in combination. Adverse effects: headache, oral irritation.
Varenicline	Nicotinic ACh receptor partial agonist. Diminishes effect on reward system, but also reduces withdrawal. Adverse effects: GI discomfort, sleep disturbance. Varen icline helps nicotine cravings decline .

Medically supervised opioid withdrawal and relapse prevention

Methadone	Long-acting oral opioid used for medically supervised opioid (eg, heroin) withdrawal or long-term maintenance therapy.
Buprenorphine	Partial opioid agonist. Sublingual form (film) used to suppress withdrawal and for maintenance therapy. Partial agonists can precipitate withdrawal symptoms in opioid-dependent individuals or when administered shortly after use of a full agonist.
Naloxone	Short-acting opioid antagonist given IM, IV, or as a nasal spray to treat acute opioid overdose, particularly to reverse respiratory and CNS depression.
Naltrexone	Long-acting oral opioid antagonist used after detoxification to prevent relapse. May help alcohol and nicotine cessation, weight loss. Use nalt rexone for the long trex back to sobriety.